

Sharma_DSC520Week3.2

2024-12-13

```
# Assignment: ASSIGNMENT 3
```

```
# Name: Sharma, Vijay
```

```
# Date: 2024-12-12
```

```
# Read CSV file for American Community Survey
```

```
surveydata = read.csv("acs-14-1yr-s0201.csv")
```

```
# There are 8 columns in the dataset and here are the details regarding  
# of data and what data type it can be in
```

```
# 1. Id - Identity column with varchar or string data type
```

```
# 2. Id1 - Int column strip out of Id Column and should be an Integer data type # 3.
```

```
# 3. Geography - This is varchar or string type column showing city and county name
```

```
# 4. PopGroupId - This is Id column and should be shown as Integer.
```

```
# 5. DisplayLabel(PopGroupId) - consider that as a string column
```

```
# 6. RacesReported - Integer Data type column.
```

```
# 7. HSDegree - Decimal Data type column
```

```
# 8. BachDegree - Decimal Data type column
```

```
str(surveydata)
```

```
## 'data.frame': 136 obs. of 8 variables:
```

```
## $ Id : chr "0500000US01073" "0500000US04013" "0500000US04019" "0500000US06001"
```

```
## $ Id2 : int 1073 4013 4019 6001 6013 6019 6029 6037 6059 6065 ...
```

```
## $ Geography : chr "Jefferson County, Alabama" "Maricopa County, Arizona" "Pima County,
```

```
## $ PopGroupID : int 1 1 1 1 1 1 1 1 1 1 ...
```

```
## $ POPGROUP.display.label: chr "Total population" "Total population" "Total population" "Total popu
```

```
## $ RacesReported : int 660793 4087191 1004516 1610921 1111339 965974 874589 10116705 314551
```

```
## $ HSDegree : num 89.1 86.8 88 86.9 88.8 73.6 74.5 77.5 84.6 80.6 ...
```

```
## $ BachDegree : num 30.5 30.2 30.8 42.8 39.7 19.7 15.4 30.3 38 20.7 ...
```

```
# nrow() function to show number of rows in surveydata
```

```
nrow(surveydata)
```

```
## [1] 136
```

```
# ncol() function to show number of columns in surveydata
```

```
ncol(surveydata)
```

```
## [1] 8
```

```
# call ggplot2 library to plot histogram with HSDegree variable
```

```
library(ggplot2)
```

```
ggplot(surveydata) +
```

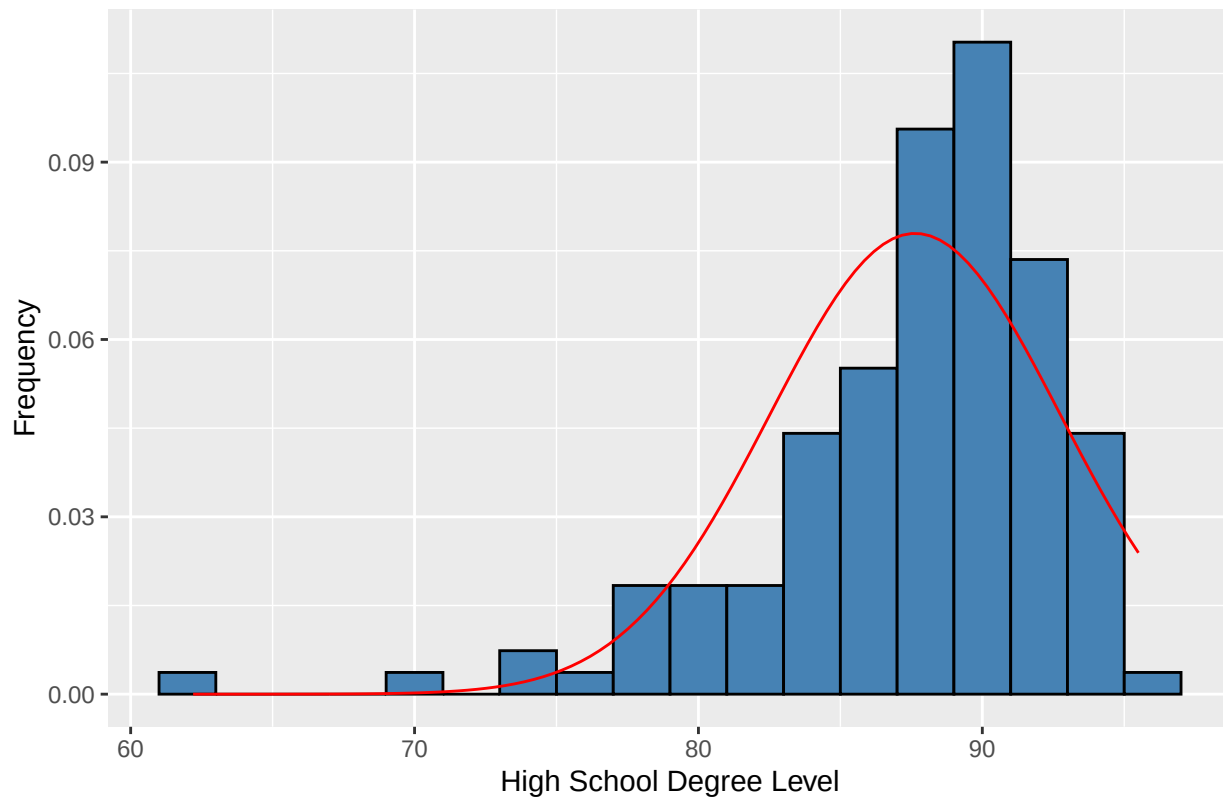
```
geom_histogram(mapping = aes(x = HSDegree, y = ..density..), fill = "steelblue", colour="black",binwidth = 5)

stat_function(fun = dnorm, args = list(mean = mean(surveydata$HSDegree), sd = sd(surveydata$HSDegree)))

labs(title = "Distribution of High School Degree" ,
      x = "High School Degree Level",
      y = "Frequency")
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Distribution of High School Degree



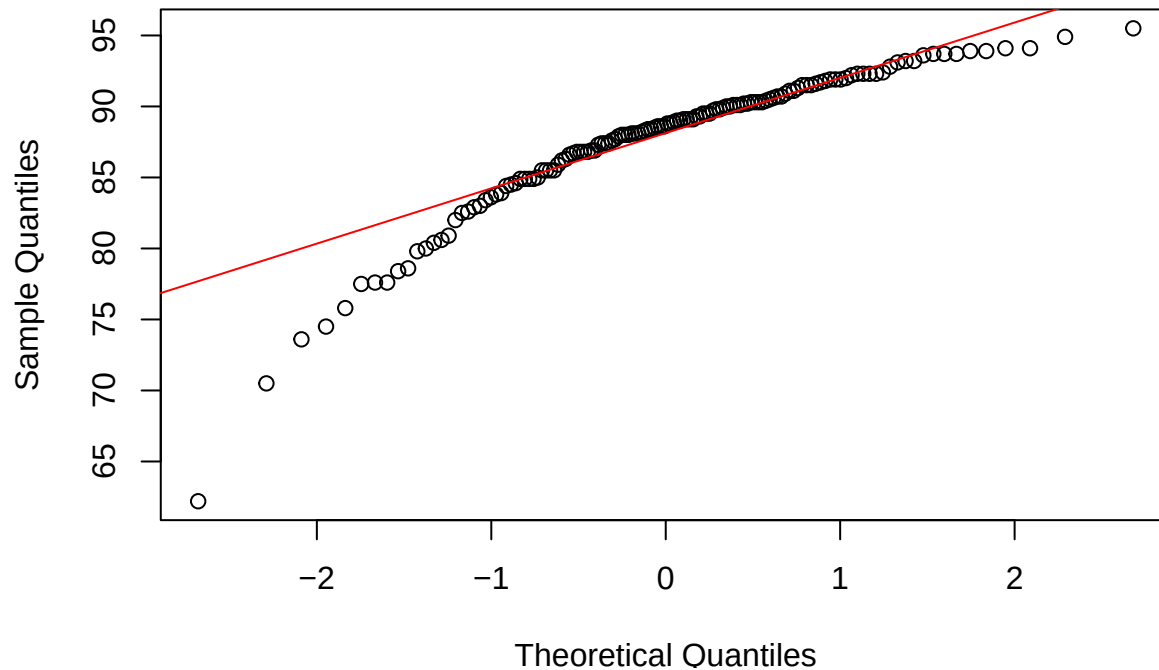
#Answers:

```
# 1. This is not Unimodel Histogram
# 2. It is even not symmetrical
# 3. It is not bell shaped Too.
# 4. and not even normal distributed
# 5. It is definitely Skewed and in the the right direction.
# 6. Based on the histogram, a normal distribution cannot be accurately used as a model for this data b
```

#Probability plot for HSDegree variable

```
qqnorm(surveydata$HSDegree)
qqline(surveydata$HSDegree,col = "red") # Adds a reference line to the plot [5, 8]
```

Normal Q-Q Plot



#Answer the following questions based on the Probability Plot

#Based on what you see in this probability plot, is the distribution approximately normal? Explain how you know.

#Answer: This is not normal distribution

#If not normal, is the distribution skewed? If so, in which direction? Explain how you know.

#Answer: Most of the data is not lying under straight line and it is mostly skewed in the middle in # p

```
library (pastecs)
stat.desc (surveydata)
```

```
##          Id          Id2 Geography PopGroupID POPGROUP.display.label
## nbr.val   NA 1.360000e+02      NA          136                NA
## nbr.null  NA 0.000000e+00      NA           0                NA
## nbr.na    NA 0.000000e+00      NA           0                NA
## min       NA 1.073000e+03      NA           1                NA
## max       NA 5.507900e+04      NA           1                NA
## range     NA 5.400600e+04      NA           0                NA
## sum       NA 3.649306e+06      NA          136                NA
## median    NA 2.611200e+04      NA           1                NA
## mean      NA 2.683313e+04      NA           1                NA
## SE.mean   NA 1.323036e+03      NA           0                NA
## CI.mean   NA 2.616557e+03      NA           0                NA
## var       NA 2.380576e+08      NA           0                NA
## std.dev   NA 1.542911e+04      NA           0                NA
## coef.var  NA 5.750024e-01      NA           0                NA
##          RacesReported      HSDegree      BachDegree
## nbr.val   1.360000e+02 1.360000e+02 136.0000000
## nbr.null  0.000000e+00 0.000000e+00  0.0000000
```

```
## nbr.na      0.000000e+00 0.000000e+00      0.000000
## min        5.002920e+05 6.220000e+01     15.400000
## max        1.011670e+07 9.550000e+01     60.300000
## range      9.616413e+06 3.330000e+01     44.900000
## sum        1.556385e+08 1.191800e+04 4822.700000
## median     8.327075e+05 8.870000e+01     34.100000
## mean       1.144401e+06 8.763235e+01     35.461029
## SE.mean    9.351028e+04 4.388598e-01      0.815452
## CI.mean    1.849346e+05 8.679296e-01      1.612714
## var        1.189207e+12 2.619332e+01     90.434988
## std.dev    1.090508e+06 5.117941e+00      9.509731
## coef.var   9.529072e-01 5.840241e-02      0.268174
```

#In this distribution it is Positively Skewed and the values are more concentrated towards the #right s

```
library(moments)
```

```
moments::skewness(surveydata$HSDegree)
```

```
## [1] -1.69341
```

#Negative skewness value indicates here that the distribution has a longer tail on the left side of the

```
moments::kurtosis(surveydata$HSDegree)
```

```
## [1] 7.462191
```

*#Kurtosis is a statistical measurement that describes the shape of a distribution's peak and tails in r
#- A positive kurtosis value indicates that the distribution has a sharper peak and lower tails than*